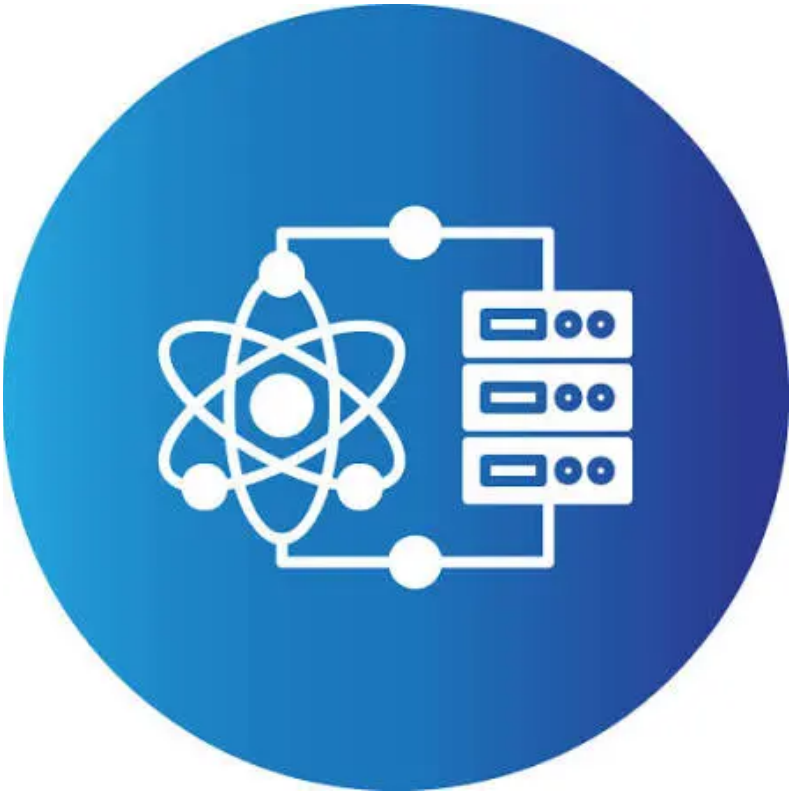


DATA SCIENCE V1.0

Lesson 4

Documentation: Data Science V1.0

Advanced Machine Learning

Objectives

- Optimize models properly
- Understand cross-validation deeply
- Build Neural Networks
- Apply hyperparameter tuning

Topics

- Cross-validation
- GridSearch
- RandomizedSearch
- Bayesian Hyperparameter Optimization
- Feature selection
- Model selection

- Neural Networks basics
- CNN fundamentals

Resources

- TensorFlow:
  - https://www.tensorflow.org/tutorials
- Keras:
  - https://keras.io/examples/
- Deep Learning Book (Goodfellow):
  - https://www.deeplearningbook.org/

Practice Project – Model Optimization Challenge
Option A: Tabular advanced modeling
Option B: Image classification (MNIST or CIFAR)

Requirements

- Build baseline model
- Apply cross-validation
- Tune at least 5 hyperparameters
- Compare results of different hypertuning methods (random search, grid search, bayesian optimization) configurations
- Plot loss function curves

Deliverables

- Performance improvement log
- Final tuned model
- Explanation:
  - Bias vs variance
  - Why selected final model?